Agentic Phishing Link Analysis: A Sandboxed-Browser Agent with an End-to-End MLOps Pipeline

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1. Problem and Objective

URL-only classifiers judge a link from its text alone. They cannot see what the page actually does, so they miss fresh domains, redirect chains, and lookalike pages. This project builds an **agent that visits the link in an isolated, sandboxed browser**, collects evidence from the live page (redirects, forms, page text, certificate), and combines it with a URL-based ML model to return a verdict (safe, suspicious, or phishing) with plain-language reasons. The agent is delivered through a complete, reproducible MLOps pipeline.

2. Agent Design

For each submitted link the agent uses a small set of tools: **(1) URL model** scores the URL features; **(2) sandboxed browser** (headless Chromium) follows redirects and records the final domain, a screenshot, visible text, iframes, and forms (password fields, forms that submit to a different domain); **(3) certificate and domain checks** compare the final domain with the original and with the brand shown on the page; **(4) reasoning step** (LLM) weighs all evidence together with the model score and outputs the verdict, confidence, and reasons.

Safety controls: the browser runs in a locked-down container with no credentials or cloud role, short timeouts, downloads disabled, observe-only behavior (no form submission), and blocked access to private network addresses. Page text is treated as untrusted data to resist prompt injection, and the ML score acts as an independent second opinion.

3. Key Features

1. **URL Feature Extraction and Model:** URL length, symbols, HTTPS usage, and domain characteristics feed the URL classifier.
2. **Sandboxed Browsing:** Safe, time-limited visit of the link in an isolated headless browser.
3. **Page Evidence Extraction:** Redirect chain, forms, iframes, page title and text, and certificate details.
4. **Agent Reasoning:** Combines URL score and page evidence into a verdict with confidence and reasons.
5. **Data Validation and Versioning:** Checks for missing, duplicate, and invalid data; datasets and preprocessing tracked with DVC on S3.
6. **Training and Experiment Tracking:** Multiple models compared and logged in MLflow, including URL-only, content-only, and combined configurations.
7. **Analysis API:** A FastAPI endpoint that accepts a URL and returns the verdict, evidence summary, and top contributing factors.
8. **Dockerization and Cloud Deployment:** Containerized API and browser sandbox deployed on AWS (ECR and ECS).
9. **CI/CD:** GitHub Actions runs tests, data validation, safety and prompt-injection tests, and container builds.
10. **Monitoring:** Verdict statistics, latency, browser failures and timeouts, and data drift with Evidently or custom metrics.

4. Models and Evaluation

Logistic Regression, Random Forest, and XGBoost are trained on URL features and on page-based features (from the PhiUSIIL dataset and a small crawled set), and an LLM reasoning layer produces the final explanation. Three configurations are compared: **URL-only, content-only, and combined agent**, using precision, recall, F1-score, and confusion matrices, plus latency and cost per analysis. Tests on recently reported phishing links and popular legitimate sites, and a set of prompt-injection pages, check real-world performance and robustness.

5. MLOps Stack

- **Language and ML:** Python, Scikit-learn, XGBoost, Pandas, NumPy, SHAP
- **Agent and Browser:** Playwright with headless Chromium; LLM via Amazon Bedrock or a similar API
- **Versioning and Tracking:** Git/GitHub, DVC (S3 remote), MLflow
- **Serving and Deployment:** FastAPI, Docker, AWS (S3, ECR, ECS); CI/CD with GitHub Actions
- **Monitoring:** Evidently AI and custom metrics

6. Expected Outcome

A reproducible, containerized phishing analysis system with a versioned dataset, tracked experiments, a cloud-deployed API, automated tests, and monitoring, in which a sandboxed browser agent investigates the link itself and explains its verdict. The evaluation shows how much page evidence improves on URL-only detection.